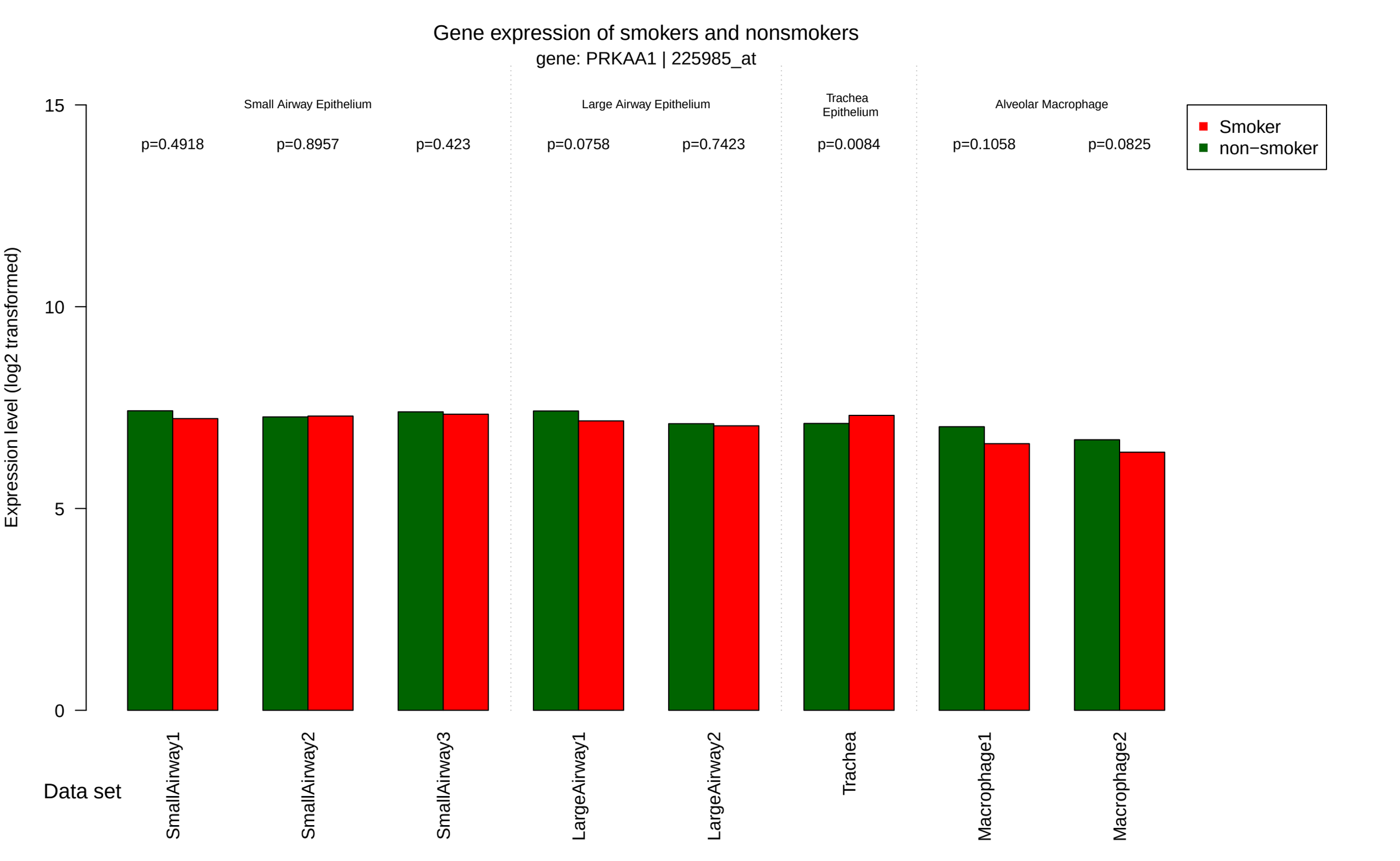
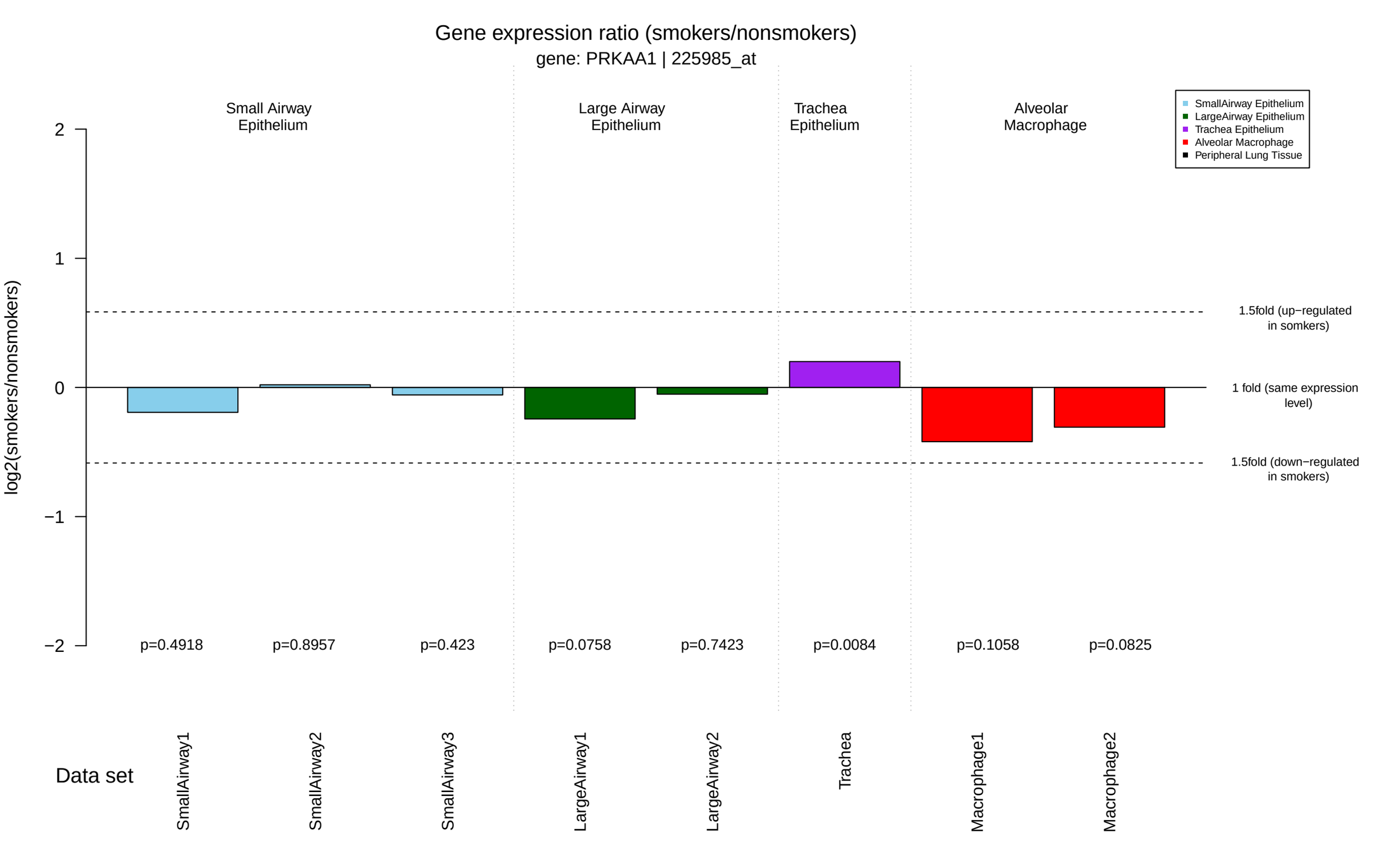
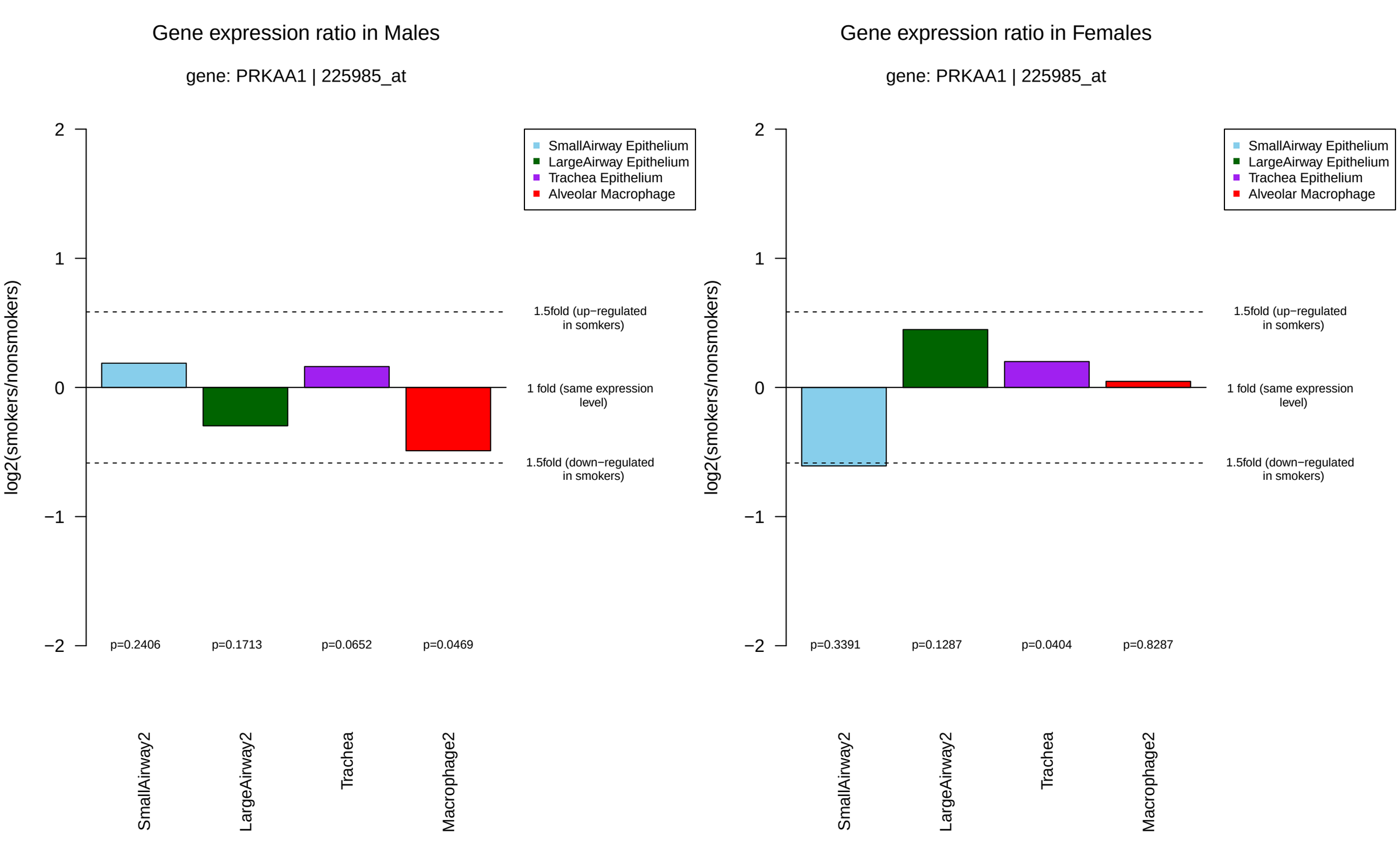
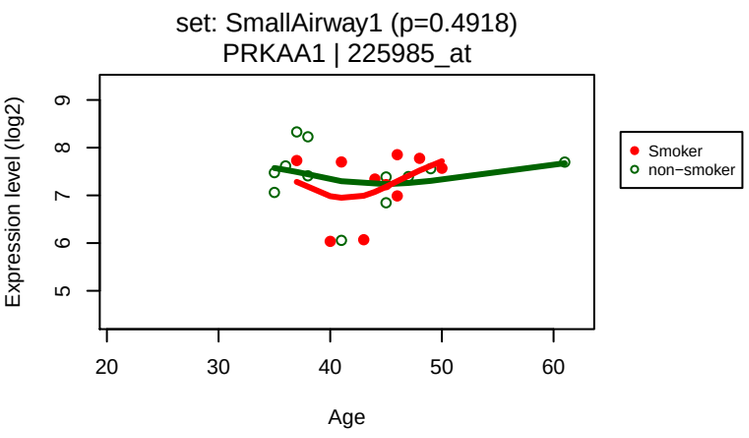
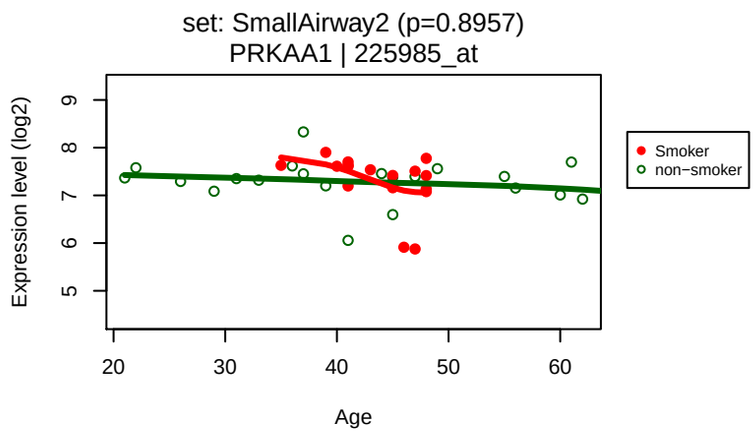
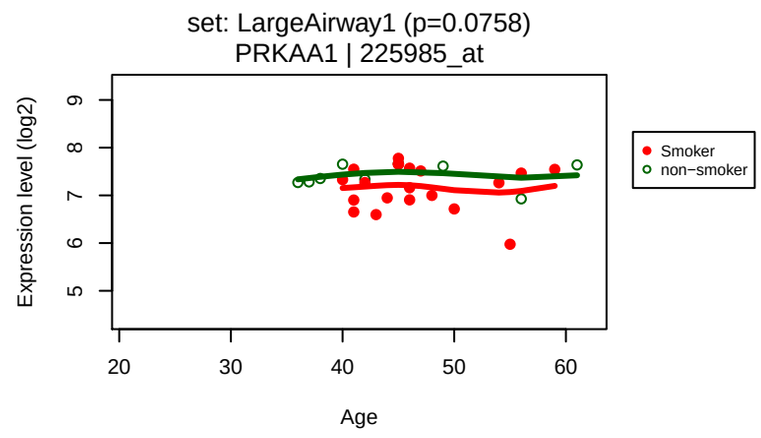

Gene expression of smokers and nonsmokers
gene: PRKAA1 | 225985_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4918
p=0.8957
p=0.423
p=0.0758
p=0.7423
p=0.0084
p=0.1058
p=0.0825
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

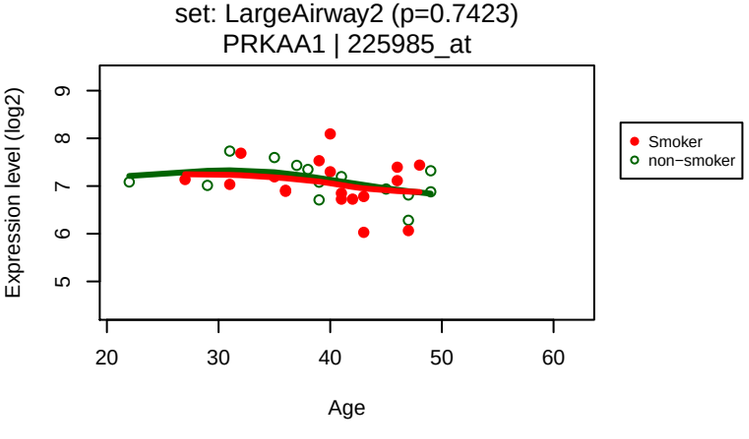

Gene expression ratio (smokers/nonsmokers)
gene: PRKAA1 | 225985_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4918
p=0.8957
p=0.423
p=0.0758
p=0.7423
p=0.0084
p=0.1058
p=0.0825
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

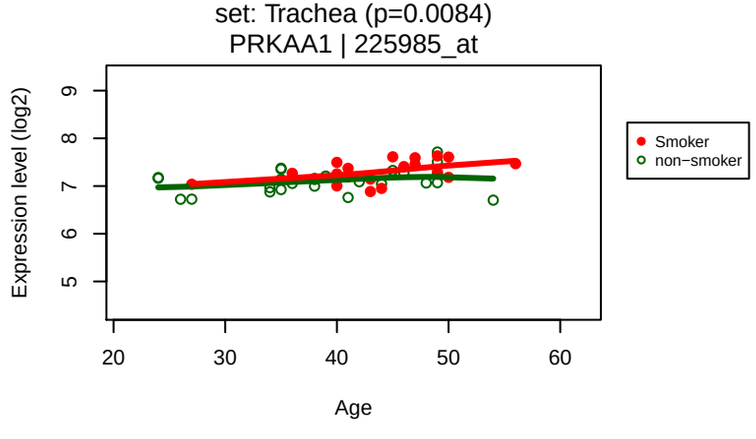

Gene expression ratio in Males
gene: PRKAA1 | 225985_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2406
p=0.1713
p=0.0652
p=0.0469
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRKAA1 | 225985_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3391
p=0.1287
p=0.0404
p=0.8287
SmallAirway2
LargeAirway2
Trachea
Macrophage2

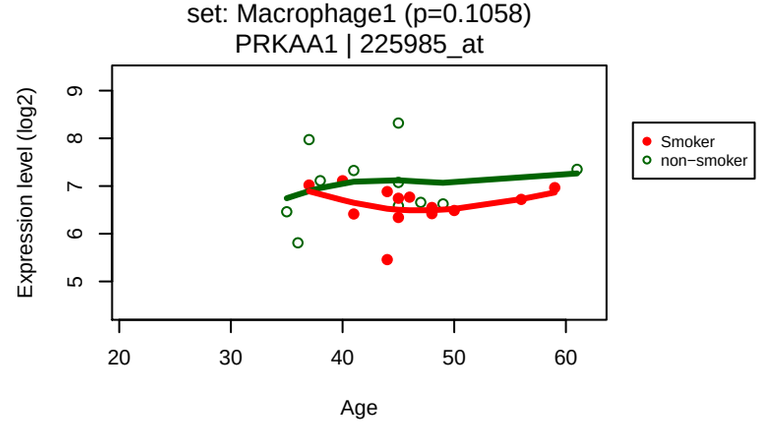

set: SmallAirway1 (p=0.4918)
PRKAA1 | 225985_at
Expression level (log2)
Age
Smoker
non−smoker

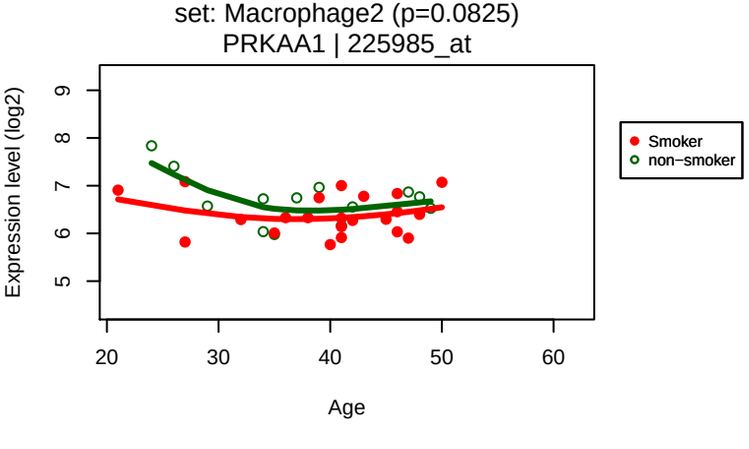

set: SmallAirway2 (p=0.8957)
PRKAA1 | 225985_at
Expression level (log2)
Age
Smoker
non−smoker

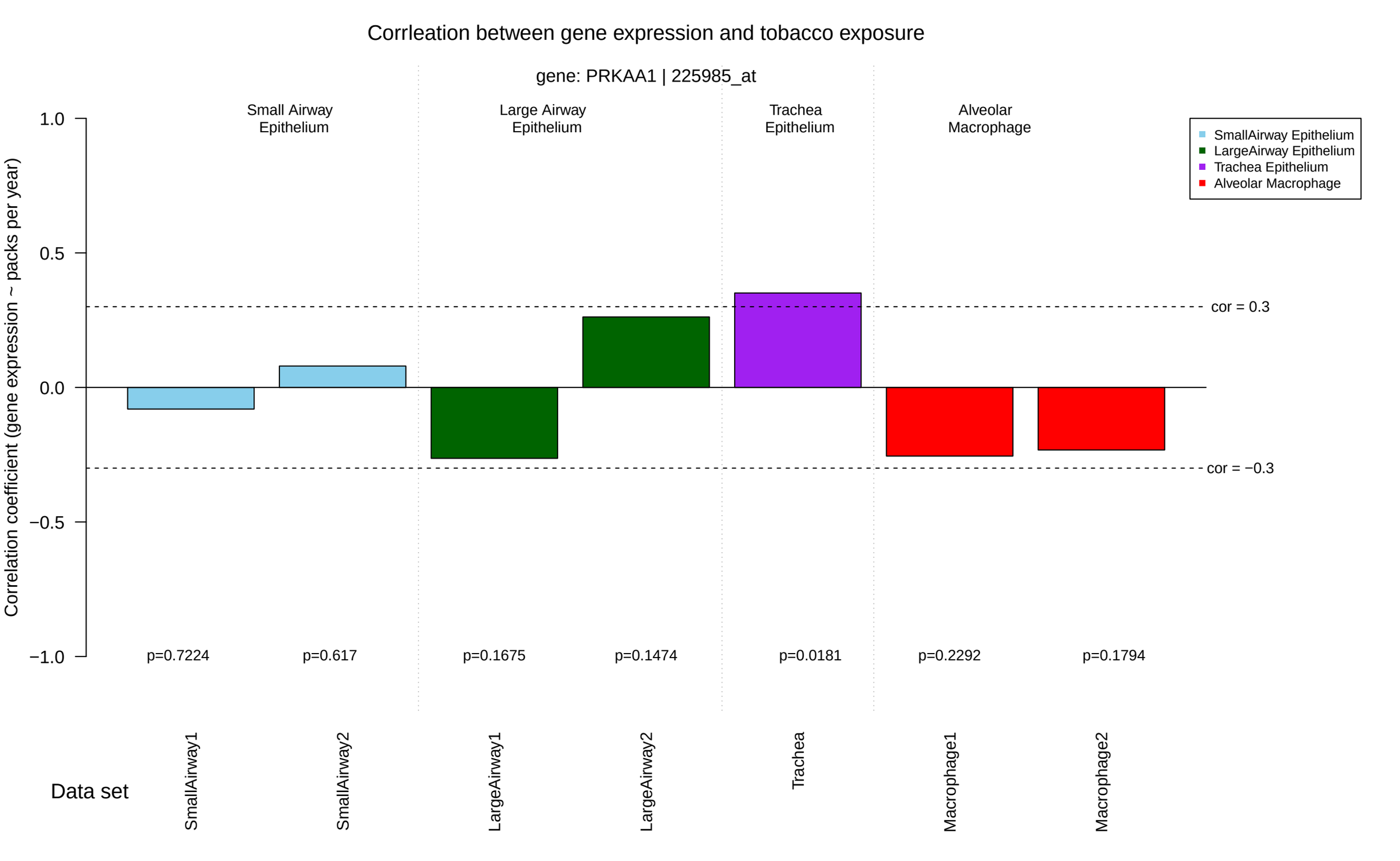

set: LargeAirway1 (p=0.0758)
PRKAA1 | 225985_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7423)
PRKAA1 | 225985_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0084)
PRKAA1 | 225985_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1058)
PRKAA1 | 225985_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0825)
PRKAA1 | 225985_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRKAA1 | 225985_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.7224
p=0.617
p=0.1675
p=0.1474
p=0.0181
p=0.2292
p=0.1794
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2